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DRAFT AS OF 7/29/2026

**AIR QUALITY BUREAU
NEW SOURCE REVIEW PERMIT
Issued under 20.2.72 NMAC**

Certified Mail No:
Return Receipt Requested

NSR Permit No: 10883
Facility Name: Yucca Growth Microgrid

Facility Owner: Yucca Growth Infrastructure, LLC.
Facility Operator: Bloom Energy Corp

Mailing Address: 600 Congress Ave, Suite 15041
Austin, TX 78701

TEMPO/IDEA ID No: 41351 - PRN20260001
AIRS No: 350134104

Permitting Action: Regular – New Construction Permit
Source Classification: TV Major

Facility Location: 341,070 m E by 3,521,500 m N, Zone 13; Datum WGS84
County: Doña Ana

Air Quality Bureau Contact Julia Kuhn
Main AQB Phone No. (505) 476-4300

Cindy Hollenberg
Bureau Chief
Air Quality Bureau

Date

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SCIENCE | INNOVATION | COLLABORATION | COMPLIANCE

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PART A FACILITY SPECIFIC REQUIREMENTS**A100 Introduction**

- A. This is a new permit.

A101 Permit Duration (expiration)

- A. The term of this permit is permanent unless withdrawn or cancelled by the Department pursuant to Section B113 and/or NMSA 1978, Section 74-2-12.B.
- B. Except for the equipment identified in Table 104.A, the construction, installation and/or operation of any stationary sources or portable stationary sources of regulated air pollutants is not authorized by this by permit and explicitly prohibited. This prohibition includes temporary, portable, and/or exempt sources pursuant to 20.2.72.202 NMAC.

A102 Facility: Description

- A. The function of the facility is to generate power to operate a data center using Bloom Energy Server System fuel cells. This system is a type of solid oxide fuel cell system (SOFC) which electrochemically converts natural gas into electricity using an electrochemical reaction.
- B. This facility is located approximately 3.6 miles South of Santa Teresa in New Mexico, Doña Ana County.
- C. This permit application is for a New Source Review (NSR) construction permit issued pursuant to 20.2.72.200.A(1) NMAC. The facility will be a microgrid power generation facility that will consist of a maximum of 2,275 solid oxide fuel cell (SOFC) clusters with a total electric generating capacity of 2,462 MW. The electric generating capacity of each module is either 260 kW or 325 kW. Fuel cells are arranged in cell stack -> power module -> energy server -> cluster -> inverter group/platform. The description of this modification is for informational purposes only and is not enforceable.
- D. Tables 102.A and Table 102.B show the total potential emission rates (PER) from this facility for information only. This is not an enforceable condition and excludes emissions from Minor NSR exempt activities per 20.2.72.202 NMAC.

Table 102.A: Total Potential Emission Rate (PER) from Entire Facility

Pollutant	Emissions (tons per year)
Nitrogen Oxides (NOx)	37.2

Table 102.A: Total Potential Emission Rate (PER) from Entire Facility

Pollutant	Emissions (tons per year)
Carbon Monoxide (CO)	161.21
Volatile Organic Compounds (VOC) ¹	124.0
Sulfur Dioxide (SO ₂)	0.08
Particulate Matter (PM) ²	75.4
Particulate Matter 10 microns or less (PM ₁₀)	75.4
Particulate Matter 2.5 microns or less (PM _{2.5})	75.4
Greenhouse Gas (GHG) as CO ₂ e ³	8,820,970

1. VOC includes emissions from Fugitives and SSM.

2. PM is a regulated new source review pollutant per 20.2.74 NMAC Prevention of Significant Deterioration. No ambient air quality standards apply to PM.

3. For reporting purposes only under 20.2.73.300 NMAC.

Table 102.B: Total Potential Emissions Rate (PER) for *Hazardous Air Pollutants (HAPs)

Pollutant	Emissions (tons per year)
Benzene	0.2
Methanol	1.2
Xylenes	0.7
Total HAPs **	1.4

* HAP emissions are already included in the VOC emission total.

** The total HAP emissions may not agree with the sum of individual HAPs because only individual HAPs greater than 1.0 tons per year are listed here.

A103 Facility: Applicable Regulations

- A. The permittee shall comply with terms, conditions and requirements specified in this permit. A list of all applicable sections of the requirements presented in Table 103.A.

Table 103.A: Applicable Requirements

Applicable Requirements	Federally Enforceable	Unit No.
20.2.1 NMAC General Provisions	X	Entire Facility
20.2.7 NMAC Excess Emissions	X (Except for Sections 6(b); 110(b)(15); 111; 112; 113; 115; and 116 that are State Enforceable Only)	Entire Facility

Table 103.A: Applicable Requirements

Applicable Requirements	Federally Enforceable	Unit No.
20.2.23 NMAC Fugitive Dust Control	X	All Construction Activities at or for the Facility
20.2.61 NMAC Smoke and Visible Emissions	X	Entire Facility
20.2.70 NMAC Operating Permits	X	Entire Facility
20.2.71 NMAC Operating Permit Emission Fees	X	Entire Facility
20.2.72 NMAC Construction Permit	X	Entire Facility
20.2.73 NMAC Notice of Intent and Emissions Inventory Requirements	X	Entire Facility
20.2.75 NMAC Construction Permit Fees	X	Entire Facility
20.2.80 NMAC Stack Heights	State Only	All Stacks

A104 Facility: Regulated Sources

A. Table 104.A lists the emission units authorized for this facility.

Table 104.A: Regulated Sources List

Unit No.	Source Description	Make	Model	Serial No.	Construction Date	Manufacture Date	Manufacturer Rated Capacity /Permitted Capacity
SOFC-1 through SOFC-2275	SOFC Clusters – 260 kW and 325 kW Units	Bloom	Energy Server 6.5	TBD	TBD	TBD	2,462 MW ²
SSM	Startup, Shutdown, and Maintenance – Venting natural gas during Centralized Desulfurization System (CDS) Filter Replacement	N/A	N/A	N/A	TBD	N/A	N/A

1. A group of 260 kW and/or 325 kW Bloom Energy Server “units” that share a stack are referred to as an “cluster” for the purpose of this permit.

2. “MW” refers to megawatts of electricity.

A105 Facility: Control Equipment

A. There is no air pollution control equipment for this facility.

A106 Facility: Allowable Emissions

A. The following Section lists the emission units and their allowable emission limits. (20.2.72.210.A and B.1 NMAC).

Table 106.A: Allowable Emissions

Unit No.	NO _x ¹ pph	NO _x ¹ tpy	CO pph	CO tpy	VOC pph	VOC tpy	SO ₂ pph	SO ₂ tpy	PM / PM ₁₀ pph	PM / PM ₁₀ tpy	PM _{2.5} pph	PM _{2.5} tpy
SOFC-1 through SOFC-2275	8.5	37.2	36.8	161.2	28.3	124.0	0.02	0.1	17.2	75.4	17.2	75.4

1 Nitrogen dioxide emissions include all oxides of nitrogen expressed as NO₂

2 Excess emissions shall be reported in accordance with 20.2.7 NMAC and Conditions B107 and B109.C.

A107 Facility: Startup, Shutdown, & Maintenance (SSM) Requirements

- A. The allowable startup, shutdown, or maintenance (SSM) activities and maximum allowable emission limits for those activities facility are listed in Section A107 and were relied upon by the Department to determine compliance with applicable regulations.

Table 107.A: Authorized SSM Units and Activities

Unit No.	Description	Make	Model	Serial No.	Construction/ Modification Date
SSM	Natural gas venting during desulfurization bed media and desulfurization filter replacement for initial start-ups, maintenance and shutdown	N/A	N/A	N/A	TBD

- B. The permittee shall limit VOC emissions from all activities related to routine or predictable startup, shutdown, and scheduled maintenance (SSM) to no more than 0.34 tons per year.
- C. The authorization of emission limits for routine or predictable startup, shutdown, or scheduled maintenance does not supersede the requirements to minimize emissions according to General Conditions B101.F and B107.A.

SSM condition option 1:

- D. Method for Determining Compliance with the Startup, Shutdown and Maintenance Natural Gas Venting Emission Limit in Section 107.B – (Unit SSM)

Requirement: The permittee shall comply with the monitoring, recordkeeping, and reporting of this condition to demonstrate compliance with the limits from SSM emissions.

- (1) The permittee shall calculate the emissions from each event using Equations 1 and 2, the maximum volume of natural gas (V_{max}) in cubic feet vented per activity, and the VOC emission factors for each activity. The permittee shall use the maximum volume of natural

gas ($V_{\max}$) in this condition when determining emissions from SSM events, even if the actual volume of gas vented was less.

Note: The maximum volume of natural gas $V_{\max}$ is the maximum amount of natural gas that can be vented from the specific activity, for example venting during one desulfurization startup event based on its physical limitations. Furthermore, the $V_{\max}$ of natural gas applies per unit of that activity and is not the sum of all natural gas vented during the event.

Equation 1: $([\text{Maximum standard cubic feet (scf)/startup}] \times [\text{Startup/month}]) + ([\text{Maximum scf/maintenance event}] \times [\text{Maintenance events/month}]) \times [\text{VOC emission factor lb/scf}] / [2,000 \text{ lb/ton}] = \text{VOC tons of emissions per month}$

Where:

Maximum scf/startup = 780,900 $V_{\max}$ vented from one startup event at one CDS unit.

Maximum scf/ maintenance = 520,600 $V_{\max}$ vented from one maintenance event at one CDS.

Startup/month = The number of total startup events that occur at all CDS units each month.

Maintenance events/month = The number of total maintenance events that occur at all CDS units each month.

VOC emission factor (lb/scf) = The sum of the emission factors for each VOC constituent in the current inlet analysis of the natural gas calculated using **Equation 2**.

Equation 2: $[\text{Mole Fraction of VOC Constituent}] \times [\text{Molecular Weight of VOC Constituent (lb/lb-mol)}] / [379.4 \text{ scf/lb-mol}]$

- (2) Each month, the permittee shall calculate the monthly rolling 12-month tpy VOC emissions.
- (3) **Emissions Due to Preventable Events:** Emission limits are 0.0 lb/hr and tons/yr for emissions due to preventable events. Emissions that are due entirely or in part to poor maintenance, careless operation, or any other preventable equipment breakdown shall be reported as excess emissions in accordance with 20.2.7 NMAC.
- (4) **Emission Limits for Unpermitted Activities:** Emission limits are 0.0 lb/hr and tons/yr for emissions for venting from equipment that have no limits in Table 107.A. Emissions due to SSM other than those represented in Section 107.B shall be reported as excess emissions in accordance with 20.2.7 NMAC.

Monitoring:

- (1) The permittee shall perform an extended gas analysis at the facility inlet at least once per year.
- (2) The permittee shall monitor each venting event, the equipment that was vented, and the cause of the event.
- (3) Each month, the permittee shall continuously monitor:
 - (a) the number of venting events during the month,
 - (b) the total calculated volume of vented gas during the month using $V_{\max}$ for the equipment that was vented as defined in the requirements of this condition,
 - (c) the total tons of VOC emissions vented during the month, and
 - (d) the monthly rolling 12-month total volume of gas and ton per year VOC emissions.
- (4) The permittee shall monitor in accordance with Condition B108 of this permit.

Recordkeeping:

- (1) For each natural gas venting event, the permittee shall keep records of:
 - (a) the gas tariff sheet documenting the %VOC,
 - (b) the total calculated volume of gas vented in scf or MMscf using the maximum volume capacity of the equipment that was vented, as defined in the requirements of this condition
 - (c) the calculations used to determine emissions
 - (d) the identification of all equipment that was vented during an event, and
 - (e) the cause or causes of the event.
 - (2) Each month, the permittee shall record the tons per month VOC emissions, and the monthly rolling 12-month tons per year (tpy) VOC emissions from SSM events.
- The permittee shall keep records in accordance with Condition B109 of this permit.

Reporting: The permittee shall report in accordance with Section B110

SSM condition option 2:

E. Method for Determining Compliance with the Startup, Shutdown and Maintenance Venting Emission Limit in Section 107.B – (Unit SSM)

Requirement: The permittee shall limit emissions from routine or predictable startup, shutdown, or maintenance to the limits in Condition A107.B.

- (1) The permittee shall process only natural gas fuel required by Condition A110.A in the Central Desulfurization System (CDS).
- (2) The permittee shall limit startup activities of all CDS units to no more than 570 vessel purges per year.
- (3) The permittee shall limit shutdown and maintenance activities of the CDS units to no more than 380 vessel purges per year.
- (4) **Emissions Due to Preventable Events:** Emission limits are 0.0 for emissions due to preventable events. Emissions that are due entirely or in part to poor maintenance, careless

operation, or any other preventable equipment breakdown shall be reported as excess emissions in accordance with 20.2.7 NMAC. (5) Emission Limits for Unpermitted Activities: Emission limits are 0.0 for emissions for venting from equipment that have no limits in Table 107.A. Emissions due to SSM other than those represented in Section 107.B shall be reported as excess emissions in accordance with 20.2.7 NMAC.
Monitoring: (1) The permittee shall monitor each SSM event, the equipment that was the source of emissions, associated equipment that contributed to the SSM event, and the cause of the event. (2) Each month, the permittee shall continuously monitor: (a) the number of CDS startups and the number of CDS shutdowns and maintenance events (b) the number of purge cycles per startup or shutdown and maintenance venting event, and (c) the number of startup or shutdown and maintenance venting events,
Recordkeeping: (1) For each SSM event, the permittee shall keep records of: (a) the natural gas fuel records required by condition A110.A, (b) a unique identifier for the CDS at which the SSM event occurred (c) the cause of the SSM event (d) the equipment associated with the SSM event (e) the number of CDS startups and the number of CDS shutdowns and maintenance events, (f) the number of purge cycles per startup or shutdown and maintenance event, (g) and the monthly rolling 12-month total number of CDS startups and the number of CDS shutdowns and maintenance.
Reporting: The permittee shall report in accordance with Section B110

F. The permittee shall report excess emissions above the authorized emission limits in Table 106.A. For equipment without emission limits, report excess emissions pursuant to Condition B110.E.

1. Initial report: the owner or operator shall file an initial report, no later than the end of the next regular business day after the time of discovery of an excess emission that includes all available information for each item in Subsection B of 20.2.7.110 NMAC.
2. Final report: the owner or operator shall file a final report that contains specific and detailed information for each item in Subsection B of 20.2.7.110 NMAC, no later than ten (10) days after the end of the excess emission.

A108 Facility: Allowable Operations

- A. This facility is authorized for continuous operation. Monitoring, recordkeeping, and reporting are not required to demonstrate compliance with continuous hours of operation.

B. Facility Throughput - Fuel Limitations

Requirement: The permittee shall limit the inlet fuel flowrate to the facility to 136,782 MMscf of natural gas per calendar year.

Pursuant to 20.2.74.300.D NMAC, any increase of the 136,782 MMscf/y of natural gas limit resulting in an emissions increase that would cause the facility to operate as or have a potential to emit of a Prevention of Significant Deterioration (PSD) major source, will result in the units approved in construction permit 10883 to be unpermitted and subject to a permit pursuant to 20.2.74.300 NMAC as if construction has not yet commenced.

Monitoring: The permittee shall continuously monitor the inlet natural gas fuel flowrate to the facility in real time using a monitoring instrument that directly measures the inlet fuel flowrate with an accuracy of $\pm 2\%$ or better as certified by the device manufacturer. The permittee shall maintain the device according to manufacturer specifications.

The permittee shall meet the General Monitoring Requirements in Section B108.

Recordkeeping: The permittee shall keep records of the monthly rolling 12-month total inlet gas flowrate, records of monitoring instrument maintenance, and records of monitoring instrument calibration and according to Section B109.

Reporting: The permittee shall report according to Section B110.

C. Facility Output - MW limitations

Requirement: The permittee shall limit the annual power output generated from natural gas from the facility to 21,567,164 MW-hr per year.

- (1) The permittee shall generate electric power only by fuel cells arranged in modules Energy Servers of 260 kW and 325 kW. This permit does not authorize any other equipment at the facility to provide power; thus, only fuel cells/modules can generate and provide power.
- (2) Fuel Cells Modules shall provide all base load and all backup power.

Pursuant to 20.2.74.300.D NMAC, any increase of the annual power output limit of 21,567,164 MW-hr per year limit resulting in an emissions increase that would cause the facility to operate as or have a potential to emit of a Prevention of Significant Deterioration (PSD) major source, will result in the units approved in construction permit 10883 to be unpermitted and subject to a permit pursuant to 20.2.74.300 NMAC as if construction has not yet commenced.

Monitoring: The Permittee shall continuously monitor and record the amount of electricity generated by the facility in units of kilowatt-hours (kWh) for the SOFC system.

The permittee shall meet the General Monitoring Requirements in Section B108

Recordkeeping: The Permittee shall keep the following records of the amount of electricity generated by the SOFC systems a on monthly rolling 12-month total. Records shall be kept according to Condition B109.

Reporting: The permittee shall report according to Section B110.

A109 Facility: Reporting Schedules

- A. The permittee shall report according to the Specific Conditions and General Conditions of this permit.

A110 Facility: Fuel and Fuel Sulfur Requirements

- A. Fuel and Fuel Sulfur Requirements

Requirement:

The permittee shall ensure that all fuel used by the SOFC 260 kW and 325 kW Units shall:

- (1) use only natural gas containing no more than 0.75 grains of total sulfur per 100 dry standard cubic, which meets the gas specifications required by the manufacturer,
- (2) process all-natural gas used by the SOFC units and clusters with the Central Desulfurization System (CDS).

Monitoring: The permittee shall continuously monitor the flowrate in real time using a monitoring instrument that directly measures the actual flowrate with an accuracy of $\pm 2\%$ or better as certified by the device manufacturer:

- (1) the actual sulfur concentration in the natural gas exiting the Central Desulfurization System (CDS). Such measurements shall be made annually on at least 5% of installed CDS by obtaining an extended gas analysis on a representative grab sample,
- (2) the CDS shall be sampled on a rotating basis such that all CDS are sampled during each rotation prior to any CDS being re-sampled.

Recordkeeping:

The permittee shall keep the following records:

- (1) a current purchase contract or tariff sheet listing the type of fuel and its quality specifications,
- (2) total measured sulfur content in the natural gas fuel exiting the CDS.

Reporting: The permittee shall report in accordance with Section B110.

A111 Facility: 20.2.61 NMAC Opacity

- A. 20.2.61 NMAC Opacity Limit (SOFC 260 kW and 325 kW Units)

Requirement: Visible emissions from all Units SOFC-1 through SOFC-2275 shall not equal or exceed an opacity of 20 percent over any 10-minute period.

Monitoring: Use of natural gas fuel constitutes compliance with 20.2.61 NMAC unless opacity equals or exceeds 20% averaged over a 10-minute period. At such time as fuel other than natural gas, or when any visible emissions are observed during steady state operation, opacity shall be measured over a 10-minute period in accordance with the procedures at 40 CFR 60, Appendix A, Method 9 as required by 20.2.61.114 NMAC. Opacity measurements shall continue on a quarterly basis per calendar year for each affected unit until such time as natural gas are used.

Recordkeeping: The opacity measures and readings shall be recorded in accordance with Method 9 in 40 CFR 60, Appendix A.

Reporting: The permittee shall report in accordance with Section B110.

A112 Facility: Haul Roads – Not Required

A113 Facility: Location Requirements

- A. The permittee shall construct the facility according to the representations in the permit application and air dispersion modeling pursuant to Section B101. The Universal Transverse Mercator (UTM) coordinates per Datum WGS84, listed in Table 113.A identify the boundaries in which the facility shall be constructed. Units SOFC-1 through SOFC-2275 must be within this area and as specified in Appendix B – Fuel Cell Stack Housing of the modeling report.

Table 113.A Facility Boundary Coordinates

Corner	Easting (m)	Northing (m)
Northwest (NW)	340,560.3	3,522,076
Northeast (NE)	341,510.2	3,522,199
Southeast (SE)	341,537.7	3,520,739
Southwest (SW)	340,570.8	3,520,974

A114 Facility: Relocation

- A. This facility or any emissions sources at this facility shall not be relocated or repositioned without permit approval pursuant to Conditions B101.A, B, C, and D and 20.2.72.202.B(4) NMAC.

A115 Alternative Operating Scenario – Not Required**A116 Compliance Plan – Not Required****A117 Pursuant to this Permit:**

- A. Pursuant to Condition B110.A “Any term or condition imposed by the department on a permit or permit revision is enforceable to the same extent as a regulation of the board.” 20.2.72.210.D NMAC The permittee shall comply with the terms and conditions of this permit, including all requirements monitoring, recordkeeping, and reporting. Failure to do so is a violation of this permit and applicable regulations.
- B. If credible evidence or information obtained by the department or provided to the department by a third party indicates that a source is not in compliance with the provisions of this permit that evidence or information may be used by the department for purposes of establishing whether a person has violated or is in violation of this permit and applicable regulations.

A200 Oil and Gas Industry – Not Required**A300 Construction Industry – Aggregate – Not Required****A400 Construction Industry – Asphalt – Not Required****A500 Construction Industry – Concrete – Not Required****POWER GENERATION INDUSTRY****A600 Power Generation Industry**

- A. This section has common equipment related to most Electric Service Operations (SIC-4911).

A601 SOFC Clusters - 260 kW and 325 kW Units

- A. SOFC Clusters – Remote Monitoring

Requirement: The permittee shall continuously monitor gas pressure and temperature within each SOFC Unit.

Each SOFC Unit shall be equipped with automated controls that initiate immediate shutdown and isolation of the affected unit following detection of out-of-range parameters. All excess emissions or emissions due to malfunction shall be calculated and reported in accordance with

20.2.7 NMAC.

Within 90 days of the issue date of permit 10883, the permittee shall submit to the Department Permit Program Manager for approval a comprehensive monitoring plan which will establish pressure and temperature ranges which indicate that atmospheric venting may be occurring. For purposes of determining compliance with the monitoring requirements of this Condition, prior to approval the permittee shall comply with the submitted monitoring plan (Condition A601.H).

The sensors shall be maintained in accordance with manufacturer specifications.

Monitoring: The permittee shall continuously monitor pressure and temperature within each SOFC Unit using sensors according to the approved monitoring plan. Any abnormal operational condition identified by the monitoring system shall be investigated by the permittee, reported and corrected in accordance with the General Reporting Requirements Section B100, and 20.2.72 NMAC Sections 210 and 212.

The permittee is subject to General Monitoring Requirements in Condition B108.

Recordkeeping: The permittee shall maintain output readings of the temperature and pressure from the sensors recorded at least once per hour.

For each out-of-range parameter, the permittee shall record:

- (1) Dates and times of onset and resolution,
- (2) Duration,
- (3) Affected unit numbers,
- (4) Cause of the temperature and/or pressure excursion,
- (5) Corrective actions taken,
- (6) Excess emissions estimated in accordance with the requirements in 20.2.7 NMAC.

The permittee shall also record the unit numbers that were shut down and shall make the monitoring plan available to Department personnel upon request.

The permittee shall maintain records in accordance with the applicable Sections in B109, B110, and B111.

Reporting: The permittee shall report in accordance with the applicable Sections B109, B110, and B111. All excess emissions or emissions due to malfunctions shall be reported in accordance with 20.2.7 NMAC.

B. 20.2.80 NMAC – Stack Heights (SOFC Cluster)

Requirement: The permittee shall ensure that each stack height of each SOFC cluster shall be at least 30 feet above ground.

Monitoring:

The permittee shall demonstrate compliance by measuring and verifying the height of each stack following installation.

The permittee shall meet the General Monitoring Requirements in Section B108

Recordkeeping: The permittee shall maintain records of the stack height measurements and in accordance with the applicable Sections in B109, B110, and B111.

Reporting: The permittee shall report in accordance with Section B110.

C. Maintenance and repair monitoring –SOFC Units and SOFC Clusters

Requirement: At all times, the permittee shall demonstrate compliance with the allowable emission limits in Table 106.A by operating and maintaining each SOFC Unit and SOFC cluster in a manner consistent with manufacturer guidelines and procedures to minimize emissions to the extent practicable.

Monitoring:

- (1) The Permittee shall install, calibrate, operate and maintain a monitoring system that produces sufficient data to ensure that manufacturer guidelines and procedures are properly followed.
- (2) At a minimum, the monitoring system shall determine when a SOFC Energy Server or SOFC cluster must be shut down for maintenance or repairs; and notify facility personnel that maintenance or repairs are needed.

The permittee shall meet the General Monitoring Requirements in Section B108

Recordkeeping:

The permittee shall maintain records in accordance with Section B109, including records of maintenance and repair activities and a copy of the manufacturer's or permittee's recommended maintenance schedule and in accordance with the applicable Sections in B109, B110, and B111.

Reporting:

The permittee shall report in accordance with Section B110.

D. Initial Compliance Testing SOFC Clusters

Requirement: The permittee shall demonstrate compliance with the allowable emission limits in Table 106.A by performing initial compliance tests for a representative number of SOFC clusters as specified under this Condition. The number of SOFC clusters subject to testing during a calendar quarter equals 1% of SOFC clusters meeting either of the following "initial stack test vesting" criteria:

- (1) the SOFC clusters have achieved their maximum normal production date at least (60) days prior to the end of such calendar quarter; OR
- (2) the SOFC clusters have started up at least (180) days prior to the end of such calendar quarter and are not part of the group of clusters that met initial stack test vesting criteria in a prior calendar quarter.
- (3) If fewer than 600 clusters meet the stack test vesting criteria above, then at least 6, or the total number of clusters meeting such criteria (whichever is lesser) shall be tested.

Monitoring: The permittee shall perform an initial compliance test in accordance with the General Testing Requirements of Section B111. Emission testing is required for NO_x, CO, VOC, PM₁₀, and

PM_{2.5}.

The permittee shall meet the General Monitoring Requirements in Section B108, except the monitoring exemptions in Condition B108.D do not apply to this condition.

Recordkeeping: The permittee shall include in each test notification and report submittal the total number of clusters satisfying the initial stack test vesting criteria of paragraph D of this Condition for that calendar quarter, and the total number of clusters to be tested during the calendar quarter, unit ID numbers, and the unit ID number and size (260 or 325 kW) of each cluster tested

The permittee shall maintain records in accordance with the applicable Sections in B109, B110, and B111.

Reporting: The permittee shall report in accordance with the applicable Sections B109, B110, and B111.

E. Periodic Testing SOFC Clusters

Requirement:

- 1) The permittee shall demonstrate compliance with the allowable emission limits in Table 106.A by completing periodic emissions tests for SOFC clusters. The permittee shall test 1% of the SOFC clusters that meet "initial stack test vesting" in each calendar year within each subsequent calendar year. The testing requirement for each "initial stack test vesting" year shall continue to apply in all subsequent years and shall not be superseded by the installation of additional clusters. The clusters that met "initial stack test vesting" year shall be tested on a rotating basis such that all clusters are sampled during each rotation prior to any cluster being retested.

The permittee shall test the lowest performing (in terms of efficiency) clusters that have not yet been tested.

- 2) If all results of initial compliance tests and the first year of periodic testing are less than 75% of the values in Table A601.E(2), or below the relevant limit of detection for the reference test method, the permittee may reduce testing to a total of 2 SOFC clusters per stack test vesting year:

Table A601.E(2)

Pollutant	Value (lb/MW-hr)
NO _x	3.00E-03
CO	1.30E-02
VOC	1.00E-02
PM ₁₀	6.08E-03
PM _{2.5}	6.08E-03

- 3) The permittee shall resume periodic annual testing of 1% of all SOFC clusters per year if a test result exceeds the relevant limit of detection for the reference test method and exceeds the values in Table A601.E(3):

Table A601.E(3)	
Pollutant	Value (lb/MW-hr)
NO _x	3.00E-02
CO	1.30E-01
VOC	1.00E-01
PM ₁₀	6.08E-02
PM _{2.5}	6.08E-02

Monitoring: The permittee shall test using EPA Reference Methods subject to the requirements and limitations of Section B108, General Monitoring Requirements. Emission testing is required for NO_x, CO, VOC, PM₁₀, and PM_{2.5} and shall be carried out as described below:

- (1) The testing shall be conducted as follows:
 - (a) Testing frequency shall be at least once per year.
 - (b) The monitoring period is defined as a calendar year.
- (2) The first periodic test shall occur for each initial stack test vesting year after completion of the last initial compliance test.
- (3) All subsequent monitoring shall occur in each succeeding monitoring period. No two monitoring events shall occur closer together in time than 25% of a monitoring period.
- (4) The permittee shall follow the General Testing Procedures of Section B111.

The permittee is subject to General Monitoring Requirements in Condition B108.

Recordkeeping: The permittee shall maintain records in accordance with the applicable Sections in B109, B110, and B111.

Reporting: The permittee shall report in accordance with the applicable Sections in B109, B110, and B111.

F. Operation and Emission Calculation

Requirement: The permittee shall demonstrate compliance with the allowable emission limits in Table 106.A by performing the following monitoring, recordkeeping, and reporting for the facility. Each Energy Server shall be equipped with a load meter, which shall continuously record the electrical output of each Energy Server.
Monitoring: The Permittee shall continuously monitor the total hourly electrical output of each Energy Server. The Permittee shall meet the General Monitoring Requirements in Condition B108.
Recordkeeping:

The Permittee shall keep records of the total hourly electrical output of each Energy Server and the total for the entire facility.

The Permittee shall use the records of the maximum electrical output per month and monthly hours of operation of each Energy Server and the emission rates from stack tests required by Conditions A601.D and E to calculate the monthly total ton per year NO_x, CO, VOC, PM₁₀, and PM_{2.5} emissions from the entire facility using the following equation:

$$E = (EF \times H) / 2,000$$

Where:

E = Monthly NO_x (or CO, VOC, PM₁₀, and PM_{2.5}) emissions (tons/month)

EF = NO_x (or CO, VOC, PM₁₀, and PM_{2.5}) emission factor (lb/MW-hr), based on the maximum results from the most recent stack tests.

H = total monthly electrical output (MWhr/mo)

The Permittee shall use the monthly records to calculate the monthly rolling 12-month total ton per year NO_x, CO, VOC, PM₁₀, and PM_{2.5} emissions from the entire facility.

Compliance with SO₂ emissions will be calculated based on the extended gas analysis results obtained annually using the following equations:

$$\text{SO}_2 \text{ emission rate (lb/MWhr)} = F / P / S * M_{\text{SO}_2} / M_S * Y$$

F = fuel consumption in standard cubic feet per hour

P = electrical capacity in operation in MW

S = standard molar volume of natural gas in ft³/lb-mol natural gas, 359 ft³/lb-mol natural gas

M_{SO₂} = molar mass of sulfur dioxide, 64 lb SO₂/lb-mol SO₂

M_S = molar mass of sulfur, 32 lb S/lb-mol S

Y = measured lb-mol S/lb-mol natural gas

$$\text{SO}_2 \text{ emissions (tons)} = \text{SO}_2 \text{ emission rate (lb/MWhr)} * H / 2000\text{-lb/ton}$$

H = total monthly electrical output (MWhr/mo)

The Permittee shall maintain records of all required calculations.

The permittee shall maintain records in accordance with the applicable Sections in B109, B110, and B111.

Reporting:

The permittee shall report in accordance with the applicable Sections in B109, B110, and B111.

G. Natural Gas Leaks

Requirement: The sum of the natural gas flow rates to all SOFC Units shall not be less than 98% of total natural gas at the inlet of the facility as measured in Condition A.108B.

Monitoring:

The permittee shall continuously monitor the sum of the natural gas flow rates to all SOFC Units using a monitoring instrument that directly measures the inlet flowrate with

an accuracy of $\pm 2\%$ or better. The permittee shall calculate the difference between the sum of natural gas flow rates to all SOFC Units and the total natural gas at the inlet of the facility as measured in Condition A.108B. The difference between these values shall be the volume of natural gas vented. The permittee is subject to General Monitoring Requirements in Condition B108.
Recordkeeping: 1. The permittee shall keep the following records: The monthly 12-month rolling sum of flowrates to all SOFC Units. 2. The monthly 12-month rolling difference between the sum of natural gas flow rates to all SOFC Units and the total natural gas at the inlet of the facility as measured in Condition A.108B.
Reporting: The permittee shall report in accordance with Section B110.

H. Continuous Monitoring System (CMS)

Within 90 days of the issue date of permit 10883, the permittee is required to submit to the Department a comprehensive monitoring plan to include at minimum continuous monitoring of the facility's CDS SSM events (A107.D and A107.E); gas flowrate (A108.B); electrical output measurements (MW) (A108.C); exiting sulfur concentration measurements (A110.A); temperature and pressure readings, (A601.A), and natural gas leaks flowrates at the SOFC Units (A601.H).

For the purpose of this condition 40 CFR 63 defines a Continuous Monitoring System (CMS) as a comprehensive term that may include, but is not limited to, continuous emission monitoring systems, continuous opacity monitoring systems, continuous parameter monitoring systems, or other manual or automatic monitoring that is used for demonstrating compliance with an applicable regulation on a continuous basis as defined by the regulation.

PART B GENERAL CONDITIONS (Attached)

**PART C MISCELLANEOUS: Supporting On-Line Documents; Definitions;
Acronyms (Attached)**